

Basic Code Analysis: ASP.NET Core Dependency Injection Lifetimes

Duration : 30 Min.

Scenario

A C# API is registering services incorrectly, causing memory leaks due to transient scopes accessing DbContext. Set proper lifecycles.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Edit 'Program.cs' to change registration for 'IDbService' from `AddTransient` to `AddScoped`.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.
6. Verify your progress by running `python run.py` locally in the terminal.

Validation

Once you have saved your files and verified your progress, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Program.cs file exists	3 Marks
TC2	IDbService registered as AddScoped	4 Marks
TC3	Transient database configurations removed	3 Marks

Total Score: 10 Marks

Important Notes

- This is an auto-evaluated question. Ensure all code edits are properly saved and the 'run.py' checks pass before submission.